

1. Evolution and History of Cloud Computing

Cloud computing did not appear suddenly – it evolved through several stages of computing technology:

- **Mainframe Computing (1950s–60s):** Large, centralized computers shared by many users via "dumb terminals." Expensive and centrally managed – the root idea of "shared computing resources."
- **Client-Server Computing (1980s–90s):** Personal computers (clients) connected to central servers. Gave users more control but reduced central resource sharing.
- **Grid Computing (1990s):** Multiple computers linked together to work as one powerful system on a shared task – introduced the idea of pooling distributed resources.
- **Utility Computing (2000s):** Computing resources offered as a metered, pay-per-use service, like electricity or water – this is the direct ancestor of cloud billing.
- **Cloud Computing (2006 onwards):** Combines virtualization, utility computing, grid computing, and web services. Amazon Web Services (AWS) launched EC2 and S3 in 2006, which is generally considered the formal start of modern cloud computing.

Easy way to remember: Mainframe → Client-Server → Grid → Utility → Cloud (resources moved from "centralized" → "distributed" → "on-demand, pay-per-use, anywhere access").

2. Introduction to Cloud Computing

Cloud computing is the delivery of computing services – servers, storage, databases, networking, software, analytics – over the **Internet ("the cloud")**, instead of owning and maintaining physical hardware.

Simple definition: "Using someone else's computers (servers), over the internet, and paying only for what you use."

NIST Definition (commonly asked):

Definition: Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, applications, services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

3. Why Cloud Computing is Becoming Highly Important

- Businesses avoid huge upfront investment in hardware (capital expenditure → operational expenditure).
- Enables **remote work** and **global collaboration** – access data/apps from anywhere.
- Supports **rapid scaling** – companies can grow or shrink IT resources instantly based on demand (e.g., e-commerce sites during sales).
- Reduces the burden of maintenance, upgrades, and IT staff for individual businesses.
- Encourages **innovation** – startups can launch products quickly without building data centers.
- Supports **big data, AI, and IoT** applications that need massive, flexible computing power. **4.**

Features of Cloud Computing

Feature	Meaning
On-Demand Self-Service	Users can provision resources (storage, servers) automatically
Broad Network Access	Services available over the internet, accessible from any device (l
Resource Pooling	Provider's resources are pooled to serve multiple customers (multi
Rapid Elasticity	Resources can scale up or down quickly based on demand.

Feature	Meaning
Measured Service	Usage is monitored and billed on a pay-per-use basis (like an ele
Multi-Tenancy	Multiple users/organizations share the same physical infrastruc

5. Cloud Computing for Various Users

- **Individual users:** Email (Gmail), storage (Google Drive, Dropbox), streaming (Netflix, Spotify).
- **Small businesses:** Website hosting, accounting software (SaaS tools) without owning servers.
- **Large enterprises:** Big data analytics, disaster recovery, global app deployment.
- **Developers:** Platforms (PaaS) to build, test, and deploy applications without managing infrastructure.
- **Government/Education:** Cost-effective large-scale data storage and e-governance services.

6. Advantages of Cloud Computing

- **Cost savings** – no need to buy/maintain physical hardware; pay-as-you-go.
- **Scalability & flexibility** – easily scale resources up/down.
- **Accessibility** – access from anywhere with internet.
- **Automatic updates** – provider handles maintenance and software updates.
- **Disaster recovery & backup** – data replicated across multiple locations.
- **Faster deployment** – new applications/services can be launched quickly.
- **Collaboration** – teams can work on shared documents/projects in real time.

7. Limitations of Cloud Computing

- **Internet dependency** – no internet means no access to cloud services.
- **Security & privacy concerns** – sensitive data is stored on third-party servers.
- **Limited control** – customers don't have full control over the underlying infrastructure.
- **Downtime risk** – if the provider's servers fail, users are affected.
- **Vendor lock-in** – difficult/costly to migrate from one cloud provider to another.
- **Ongoing costs** – long-term subscription costs may exceed one-time hardware purchase in some cases.

UNIT II: Cloud Models and Types of SaaS (6 Hours)

1. The NIST Cloud Computing Model

NIST defines cloud computing using **5 essential characteristics**, **3 service models**, and **4 deployment models** (memorize as **5-3-4 model**):

- **5 Characteristics:** On-demand self-service, Broad network access, Resource pooling, Rapid elasticity, Measured service.
- **3 Service Models:** IaaS, PaaS, SaaS.
- **4 Deployment Models:** Public, Private, Hybrid, Community.

2. Cloud Cube Model

Developed by the Jericho Forum, it classifies clouds along **four dimensions** to help organizations choose the right cloud type:

- **Internal / External** – Is the data stored inside or outside the organization's own boundary?
- **Proprietary / Open** – Is the technology owned/controlled by one vendor, or based on open, interoperable standards?
- **Perimeterised / De-perimeterised** – Does the organization operate within a traditional network boundary/firewall, or has the boundary dissolved (data accessible from outside)?
- **Insourced / Outsourced** – Is the service run by the organization's own staff, or by a third-party provider?

Purpose: Helps businesses assess security and operational risk before choosing a cloud strategy. **3.**

Deployment Models

Model	Description	Example Use Case
Public Cloud		Public cloud providers; resources shared among multiple users over the internet. AWS, Microsoft Azure, Google Cloud.
Private Cloud		Cloud exclusively for one organization; can be hosted on-premises or in a third-party data center but not shared. Banks, government agencies, etc.
Hybrid Cloud		Combination of public and private clouds, allowing data and applications to move between them.

		move between them. Companies keeping te but using public cl
Community Cloud		organizations with common concerns e, jurisdiction). Government bodies, ums

4. Service Models

Model	What It Provides	User Manages	Provider Manages	Example
	structure as a Service) age, networks OS,		Physical hardware, virtua	
				form as a Service) /run applications d data OS, servers, e Google App Engine,
	ware as a Service) are applications Just			platform, infra) Gmail, sforce

Easy memory trick: Think of a **pizza-as-a-service analogy**:

- **On-premises (traditional IT):** You make pizza at home (you manage everything).
- **IaaS:** Pizza kitchen rented – you bring ingredients and cook.
- **PaaS:** Pizza delivered half-made – you just add toppings.
- **SaaS:** Pizza delivered ready to eat – you just consume it.

5. Layers and Types of Cloud

The cloud architecture is typically layered:

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- **Infrastructure Layer** (physical servers, storage, networking – base layer)
- **Platform Layer** (OS, runtime environments, development tools)
- **Application Layer** (end-user software/applications, topmost layer)

6. Components of Cloud Computing

- **Clients** – devices (desktops, laptops, mobiles) used to access cloud services.
- **Datacenters** – collection of servers that host applications and store data.
- **Distributed Servers** – servers located in different geographic locations for redundancy and fast access.
- **Cloud-based Delivery** (SaaS, PaaS, IaaS) – the services offered.

7. Cloud Computing Service Providers

Major players offering cloud services: **Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform**

(GCP), IBM Cloud, Oracle Cloud, Salesforce.

8. Software as a Service (SaaS)

Definition: A software distribution model where applications are hosted by a provider and made available to customers over the internet, typically through a web browser – no installation needed.

9. Evolution of SaaS

- Started with **Application Service Providers (ASPs)** in the late 1990s, who hosted software for businesses.
- Evolved with the growth of the internet and web technologies into modern **multi-tenant SaaS applications** (e.g., Salesforce, launched in 1999, is considered a SaaS pioneer).
- Broadband growth, virtualization, and web 2.0 accelerated SaaS adoption in the 2000s–2010s.

10. SaaS Unification Technologies

Technologies/approaches that help combine (unify) data and functionality from multiple SaaS applications so they work together seamlessly, such as **APIs, middleware, and integration platforms**.

11. SaaS Integration Products and Technologies

Tools used to connect SaaS applications with each other or with on-premises systems:

- **API-based integration** (most common)
- **Enterprise Service Bus (ESB)**
- **iPaaS (Integration Platform as a Service)** – e.g., MuleSoft, Dell Boomi, Zapier

12. SaaS Product Selection Criteria

When choosing a SaaS product, organizations evaluate:

- **Functionality** – does it meet business requirements?
- **Security & compliance** – data protection standards.
- **Integration capability** – can it connect with existing systems?
- **Scalability** – can it grow with the business?
- **Cost** – subscription pricing vs. value delivered.
- **Vendor reliability** – uptime guarantees (SLA), support quality.
- **Customization** – flexibility to adapt to specific workflows.

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13. SaaS Integration Services

Professional/managed services that help businesses connect their SaaS applications with other systems (data migration, workflow automation, API management) – often provided by third-party integration specialists or the SaaS vendor itself.

14. Advantages of SaaS

- No installation or maintenance needed by the user.
- Accessible from any device with a browser and internet.
- Automatic updates and patches handled by the vendor.
- Lower upfront cost – subscription-based pricing.
- Easy scalability – add/remove users as needed.
- Faster deployment compared to traditional software.

UNIT III: Platform as a Service (PaaS) (6 Hours)

1. Introduction to PaaS

PaaS provides a **complete development and deployment environment in the cloud** – including servers, storage, OS, and middleware – so developers can build, test, and deploy applications **without managing the underlying infrastructure**.

Simple definition: "A ready-made platform where developers just write code – everything else (servers, OS, database engine) is managed by the provider."

2. Evolution of PaaS

- Emerged as an extension of IaaS once developers wanted to focus purely on **coding**, not infrastructure setup.
- Early PaaS offerings (mid-2000s) like Google App Engine (2008) and Force.com (Salesforce's platform) allowed developers to build apps without provisioning servers.

- Modern PaaS supports multiple languages, frameworks, containers, and microservices architectures.

3. PaaS Service Providers

Provider	Description
Acquia Cloud	PaaS specifically built for hosting and managing Drupal-based website
Amazon AWS (Elastic Beanstalk)	AWS's PaaS offering – automatically handles deployment, capacity
Google App Engine	Fully managed platform for building and deploying scalable web app
Force.com	Salesforce's PaaS for building business applications on top of the S

4. Amazon Elastic Beanstalk (detail)

- A PaaS from AWS that automates deployment: developer uploads code, and Beanstalk manages the deployment, capacity provisioning, load balancing, auto-scaling, and application health monitoring.
- Supports multiple languages: Java, .NET, PHP, Node.js, Python, Ruby, Go.
- Developer still has access to underlying AWS resources if needed (more flexible than pure PaaS).

5. Google App Engine (detail)

- Fully managed **serverless** platform.
- Automatically scales applications up/down based on traffic.
- Supports standard environment (sandboxed, specific languages) and flexible environment (custom Docker containers).
- No server management required at all.

6. Force.com (detail)

- Platform provided by Salesforce for building custom business applications.
- Uses **Apex** (proprietary programming language) and **Visualforce** for UI development.
- Deep integration with Salesforce CRM data and workflows.

7. PaaS Application Framework

A set of pre-built tools, libraries, and components that provide a foundation for building applications on the PaaS – includes:

- **Programming language support** (Java, Python, Ruby, etc.)

- **Database management tools**
- **Built-in APIs** for common functionality (authentication, messaging)
- **Application lifecycle management** (build, test, deploy)

8. PaaS Operator Verbs

Actions performed by the **operator/administrator** managing the PaaS platform itself:

- **Provision** – allocate resources.
- **Manage/Monitor** – track platform health and performance.
- **Migrate** – move applications/data between environments.
- **Secure** – configure access controls and security policies.

9. PaaS Developer Verbs

Actions performed by **developers** using the PaaS to build applications:

- **Develop/Build** – write application code.
- **Deploy** – publish the application to the platform.
- **Test/Debug** – verify application functionality.
- **Scale** – configure how the application should handle varying load (often automated).

(Easy way to differentiate: Operator Verbs = platform-level actions by admins; Developer Verbs = application-level actions by coders.)

10. Advantages of PaaS

- Faster development – developers focus only on code, not infrastructure.
- Reduced coding time using pre-built frameworks and tools.
- Automatic scalability and load balancing.
- Supports collaborative development (multiple developers working simultaneously).
- Lower cost – no need to buy/manage servers or OS licenses.
- Simplified deployment process.

11. Challenges of PaaS

- **Vendor lock-in** – applications built on one PaaS may not easily migrate to another.
- **Limited control** over underlying infrastructure and configurations.
- **Runtime/language restrictions** – must use supported languages/frameworks.
- **Security concerns** – dependent on provider's security practices.
- **Integration issues** with legacy/on-premises systems.

UNIT IV: Infrastructure as a Service (IaaS) and Cloud Data (6 Hours)

Part A: Infrastructure as a Service (IaaS)

1. Introduction & Evolution of IaaS

IaaS provides **virtualized computing resources over the internet** – virtual machines, storage, networks – allowing customers to rent infrastructure instead of buying physical hardware.

- Evolved from traditional **data center hosting/colocation** services, combined with **virtualization technology**, which allowed a single physical server to be divided into multiple virtual servers.
- Amazon EC2 (2006) is widely considered the pioneer of modern IaaS.

Simple definition: "Renting virtual computers (servers), storage, and networking on-demand, paying only for what is used – like renting an empty apartment where you bring your own furniture (OS and software)."

2. IaaS Architecture

Key layers in IaaS architecture:

- **Physical Hardware Layer** – actual servers, storage devices, network hardware in the data center.
- **Virtualization Layer (Hypervisor)** – software that creates and manages virtual machines (VMs) on top of physical hardware.
- **Virtual Resource Layer** – the VMs, virtual storage, and virtual networks that customers actually use.
- **Management/Orchestration Layer** – tools for provisioning, monitoring, scaling, and billing resources. •

User Access Layer – APIs, dashboards, and CLI tools customers use to manage their infrastructure. **3.**

SAN Model (Storage Area Network)

- A SAN is a **dedicated high-speed network** that provides block-level network access to storage, separate from the regular local area network (LAN).
- Used in IaaS to give servers fast, reliable, and scalable access to shared storage resources. •

Improves storage utilization, enables centralized storage management, and supports disaster recovery. **4.**

IaaS Providers

Major IaaS providers: **Amazon Web Services (AWS EC2), Microsoft Azure, Google Compute Engine, IBM Cloud, Rackspace, DigitalOcean.**

5. Advantages of IaaS

- **Cost-efficient** – no capital expenditure on physical hardware.
- **Scalability** – resources scale up/down instantly based on demand.
- **Flexibility** – full control over OS, storage, and installed applications.
- **Faster provisioning** – new servers can be deployed within minutes.
- **Disaster recovery** – easier backup and replication across regions.
- **Reduced maintenance burden** – provider maintains the physical hardware.

6. Disadvantages of IaaS

- **Security responsibility shared** – customer must still secure OS/applications (shared responsibility model).

- **Requires technical expertise** to manage servers, OS, and networking.
- **Ongoing operational costs** can add up over long-term usage.
- **Potential downtime** if the provider experiences outages.
- **Dependency on internet connectivity.**

Part B: Data in Cloud

7. Evolution of Network Storage in Cloud

- Traditional storage: **Direct Attached Storage (DAS)** – storage physically attached to a single server.
- Evolved to **Network Attached Storage (NAS)** – storage accessible over a network by multiple users.
- Further evolved to **Storage Area Networks (SAN)** – dedicated high-speed storage networks.
- Cloud storage represents the latest evolution: **virtualized, distributed, and globally accessible storage** delivered as a service over the internet.

8. Data as a Service (DaaS)

- A service model where **data** (not software or infrastructure) is provided on-demand to users, regardless of their location, through the cloud.
- Enables organizations to access high-quality, up-to-date data without managing the underlying data infrastructure.
- Example: Financial market data feeds delivered via cloud APIs.

9. Database as a Service (DBaaS)

- A cloud service that provides **access to a database without the need to set up physical hardware, install software, or manage the database configuration**.
- Provider handles maintenance, backups, patching, and scaling.
- Examples: Amazon RDS, Google Cloud SQL, Azure SQL Database.

10. Cloud Based Data Storage

Storage services where data is stored on remote servers accessed via the internet, managed by a cloud provider:

- **Object Storage** – for unstructured data (e.g., Amazon S3).
- **Block Storage** – for structured data needing high performance (e.g., Amazon EBS).
- **File Storage** – for shared file systems accessible by multiple instances (e.g., Amazon EFS).

11. Advantages of Cloud-Based Storage

- Accessible from anywhere with an internet connection.
- Automatic backup and data redundancy across locations.
- Highly scalable – pay only for storage used.
- Easy sharing and collaboration on files.
- Reduced hardware maintenance costs.

12. Limitations of Cloud-Based Storage Solutions

- Dependent on internet connectivity for access.
- Security and privacy risks (data stored off-premises).
- Potential latency issues for very large files.

- Ongoing subscription costs.
- Data transfer/egress fees when moving large amounts of data out of the

cloud. 13. Cloud-Based Data Storage Service Providers

Examples: **Amazon S3, Google Cloud Storage, Microsoft Azure Blob Storage, Dropbox, Box.**

UNIT V: Virtualization (6 Hours)

1. Introduction to Virtualization

Virtualization is the technology that allows creation of a **virtual (rather than physical) version** of something – such as a server, storage device, network, or operating system – enabling multiple virtual systems to run on a single physical machine.

Simple definition: "Dividing one physical computer into multiple 'pretend' computers, each behaving like an independent machine."

Virtualization is the **foundation technology that makes cloud computing possible** – it allows cloud providers to share physical hardware among many customers safely and efficiently.

2. Technical Evolution and History of Virtualization

- **1960s:** IBM introduced virtualization concepts on mainframes (CP/CMS system) to allow multiple users to share expensive mainframe hardware.
- **1990s–2000s:** x86 virtualization became practical with software like VMware, enabling virtualization on standard desktop/server hardware.
- **2000s–present:** Virtualization became central to cloud computing, enabling providers like AWS and Azure to offer virtual machines to millions of customers on shared physical infrastructure.

3. Types of Virtual Machines

Type	Description
System Virtual Machine	Provides a complete virtual environment that can run a full operating system.
Process Virtual Machine (Application VM)	Runs a single program/process in a platform-independent environment.

4. Advantages of Virtualization

- **Better hardware utilization** – multiple VMs run on one physical server, reducing waste.
- **Cost savings** – fewer physical machines needed, reducing hardware and energy costs.
- **Isolation** – a problem in one VM doesn't affect others on the same host.
- **Easy backup and disaster recovery** – VMs can be easily copied, backed up, or moved.
- **Scalability** – new VMs can be created quickly to meet demand.
- **Testing and development flexibility** – developers can create isolated test environments easily.

5. Components of Virtualization

- **Host Machine** – the physical computer on which virtualization takes place.
- **Hypervisor (Virtual Machine Monitor - VMM)** – software layer that creates and manages virtual machines, allocating physical resources (CPU, memory, storage) to each VM.
- **Guest Machine (Virtual Machine)** – the virtual system running on top of the host, behaving like an independent computer.
- **Virtual Hardware** – the virtualized CPU, memory, storage, and network interfaces presented to each guest VM.

6. Types of Virtualization

Type	Description	Example
Hardware/Server Virtualization	Creating multiple virtual servers on one physical server using a hypervisor. VMware ESXi, Microsoft Hyper-V.	

Type	Description	Example
OS-Level Virtualization	Running multiple isolated user-space instances on a single operating system kernel.	Docker, LXC

Network Virtualization		network resources into a single, administrative entity. VLANs, SDN (Software Defined Network)
Storage Virtualization		Storage from multiple devices into what appears to be a single storage unit. SAN-based storage
Desktop Virtualization		Operating system/environment as a virtual desktop accessible remotely. VDI (Virtual Desktop Infrastructure)
Application Virtualization		Application in an isolated environment separate from the OS. Application streaming, Java Virtual Machine

7. Types of Hypervisors (commonly asked alongside this topic)

- **Type 1 (Bare-Metal Hypervisor):** Runs directly on the physical hardware, without a host OS. More efficient and used in enterprise/cloud data centers. Examples: VMware ESXi, Microsoft Hyper-V, Xen.
- **Type 2 (Hosted Hypervisor):** Runs on top of a conventional host operating system, like any other application. Easier to set up but less efficient. Examples: VMware Workstation, Oracle VirtualBox.

Quick Revision Summary Table

Unit	Core Topic	Key Terms to Remember
I	Introduction to Cloud Computing	Evolution, NIST definition, features (on-demand, self-service, etc.)
II	Cloud Models & SaaS	NIST 5-3-4 model, Cloud Cube Model, deployment models
III	PaaS	App frameworks, providers (Beanstalk, etc.)
IV	IaaS & Cloud Data	IaaS architecture, SAN, DaaS, DBaaS, cloud storage
V	Virtualization	Hypervisor, host/guest, types of virtualization

Notes compiled from: DSE-201(B)T-CA – Fundamentals of Cloud Computing, SYBCA Semester III Syllabus (2025 Pattern), Department of BCA, PCCOE&R, Ravet.

